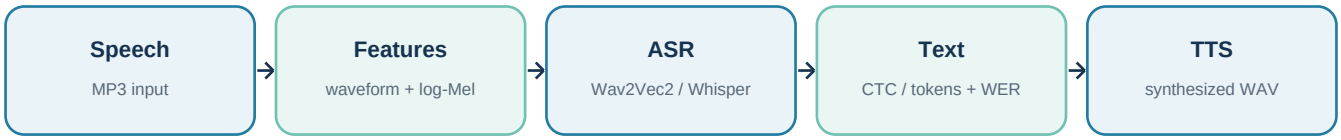
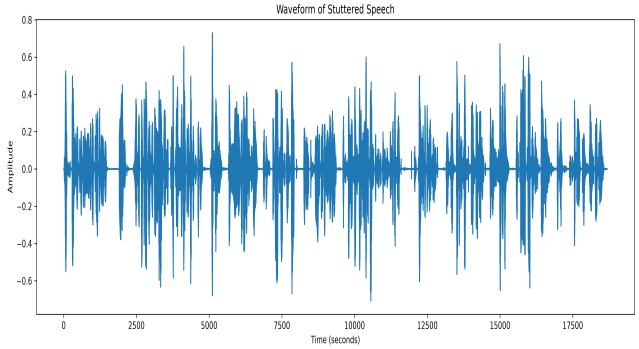
From Voice to Text and Back

A two-page study of speech features, ASR, CTC, word error rate, and speech synthesis

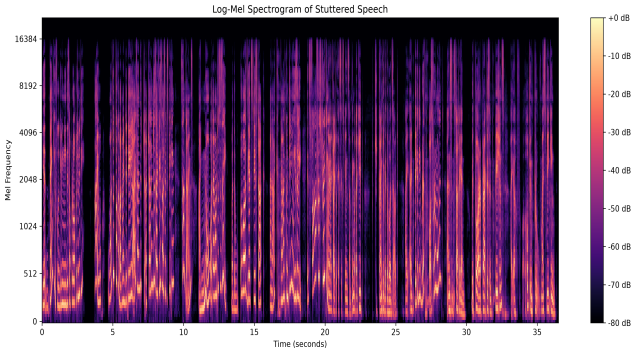


1. Objective and audio analysis

This educational project demonstrates a complete speech technology pipeline using an English recording with speech disfluencies. It is a technical demonstration—not a clinical assessment or diagnostic tool. The source recording is approximately **36.49 seconds** at **44.1 kHz**; ASR input is mono audio resampled to **16 kHz** for Wav2Vec2.



Waveform: amplitude over time



Log-Mel spectrum: 128 bands; FFT 2048; hop 512

The log-Mel representation divides the waveform into overlapping short-time frames, maps spectral energy onto 128 Mel bands, and converts power to a decibel scale. Pauses or repeated acoustic patterns can be inspected in context, but visual features alone cannot establish a stutter or identify its cause.

2. ASR models and CTC decoding

Wav2Vec2-base-960h encodes waveform samples with a convolutional feature encoder and Transformer context network, then predicts character logits over time for CTC decoding. **Whisper-base** encodes log-Mel audio features and generates text tokens with an encoder-decoder Transformer. Whisper does not use CTC; these are alternative model paths.

Connectionist Temporal Classification (CTC) handles unknown frame-to-character alignment. Greedy decoding first merges adjacent repeated tokens and then removes blank tokens. For example, A, A, blank, A, B, B becomes AAB: the blank preserves the second A.

3. ASR comparison and word error rate

Both recognizers process the same recording, but use different model architectures and decoding strategies. The following results use the current 81-word reference transcript.

Model	Decoding	Hits	Sub.	Del.	Ins.	WER
Wav2Vec2 CTC	CTC greedy	74	7	0	8	18.52%
Whisper	Encoder-decoder	81	0	0	1	1.23%

WER = (substitutions + deletions + insertions) / reference words. With 81 reference words: Wav2Vec2 = (7 + 0 + 8) / 81 = **18.52%**; Whisper = (0 + 0 + 1) / 81 = **1.23%**.

Important evaluation limitation. The 81-word reference has not been independently verified and closely follows the Whisper transcript. This biases the comparison in Whisper's favor. Treat these WER scores as provisional examples, not final accuracy claims. A human transcript checked against the audio—including a consistent policy for disfluencies—is needed for a valid comparison.

In this run, Wav2Vec2 has seven substitutions and eight insertions, while Whisper has one insertion and no substitutions or deletions against this reference. This describes only this recording and reference; it is not evidence of general model superiority.

4. Text-to-speech and discussion

The Whisper transcript is passed to Windows System.Speech and synthesized as a WAV file. This completes the return path from recognized text to speech. The generated voice is synthetic: it is not the original speaker and does not preserve the original disfluencies.

5. Findings and limitations

The project produces waveform and log-Mel visualizations, two ASR transcripts, a CTC demonstration, a model comparison with substitution/deletion/insertion counts, and a TTS output. For this single recording, the provisional calculation favors Whisper. However, because the reference is not independent, that result cannot establish that Whisper is generally more accurate. Speech models may normalize, omit, or hallucinate disfluencies; WER also depends on transcription conventions, punctuation normalization, and the quality of the reference.

Reproducibility

The supplied scripts run audio visualization, Wav2Vec2 ASR, Whisper ASR, WER, and TTS as separate steps. The Jupyter/Colab notebook combines the workflow. The architecture diagram and timed demo script are provided in the report folder. Model downloads and notebook gTTS may require internet access; Windows System.Speech is used by the PowerShell TTS script.

Conclusion

This activity connects acoustic feature extraction to recognition and synthesis while contrasting CTC-based decoding with autoregressive token generation. The next essential research step is to prepare an independently verified verbatim transcript, rerun both recognizers under consistent normalization, and recalculate WER before making accuracy claims.

Input	36.49-second English MP3, 44.1 kHz source; mono ASR
Wav2Vec2	facebook/wav2vec2-base-960h; CTC greedy decode
Whisper	base; CPU int8; English; beam size 5
Features	128 Mel bands; FFT 2048; hop 512
TTS	Windows System.Speech; synthetic output WAV

Privacy and interpretation: the recording and derived speech files require appropriate speaker permission before redistribution. This work demonstrates ASR/TTS behavior; it does not diagnose or assess a speech disorder.